INCH-POUND

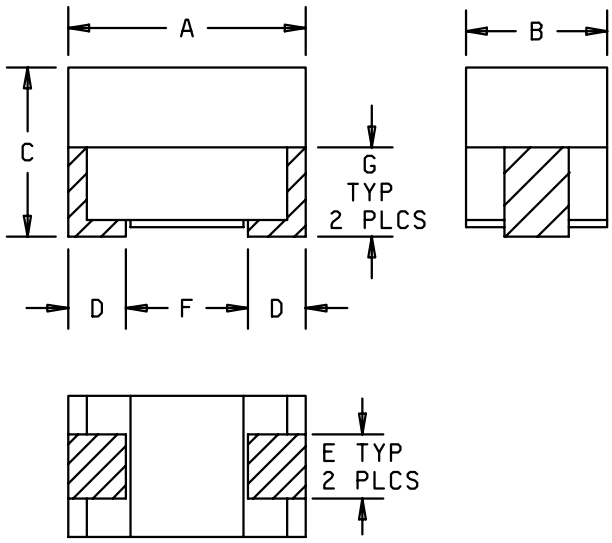
MIL-PRF-83446/34A
8 September 2005
SUPERSEDING
MIL-PRF-83446/34
1 June 2000

PERFORMANCE SPECIFICATION SHEET

# COILS, RADIO FREQUENCY, CHIP, MAGNETICALLY SHIELDED, IRON CORE-IRON SLEEVE, AND IRON CORE-FERRITE SLEEVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-83446.




| | Inches | Millimeters |
|---|---|---|
| A | .300 to .325 | 7.62 to 8.26 |
| B | .105 to .125 | 2.67 to 3.18 |
| C | .125 to .145 | 3.18 to 3.68 |
| D | .020 Min. | .508 Min. |
| E | .040 to .060 | 1.02 to 1.52 |
| F | .190 Ref. | 4.83 Ref. |
| G | .070 Ref. | 1.78 Ref. |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.



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TABLE I. Electrical characteristics (initial).

| Dash Number | Inductance µH ±10% | Q (min) | Test Frequency (L and Q) MHz | Self Resonant Frequency MHz (min) | DC Resistance Ohms (max) | Current Rating mA (max) | Incremental current Dc (mA) | Material |
|---|---|---|---|---|---|---|---|---|
| -01 | .10 | 45 | 25 | 1/ 490 | .10 | 670 | 670 | |
| -02 | .12 | 45 | 25 | 1/ 430 | .11 | 635 | 635 | |
| -03 | .15 | 45 | 25 | 1/ 415 | .12 | 610 | 610 | |
| -04 | .18 | 45 | 25 | 1/ 375 | .13 | 585 | 585 | |
| -05 | .22 | 45 | 25 | 1/ 330 | .15 | 545 | 545 | |
| -06 | .27 | 45 | 25 | 1/ 300 | .16 | 530 | 530 | |
| -07 | .33 | 44 | 25 | 1/ 260 | .18 | 495 | 495 | |
| -08 | .39 | 42 | 25 | 230 | .19 | 485 | 485 | |
| -09 | .47 | 41 | 25 | 220 | .21 | 460 | 460 | |
| -10 | .56 | 41 | 25 | 210 | .23 | 440 | 440 | |
| -11 | .68 | 39 | 25 | 180 | .24 | 430 | 430 | |
| -12 | .82 | 38 | 25 | 165 | .27 | 405 | 405 | |
| -13 | 1.00 | 37 | 25 | 150 | .30 | 385 | 385 | 2/ |
| -14 | 1.20 | 40 | 7.9 | 130 | .73 | 247 | 247 | |
| -15 | 1.50 | 41 | 7.9 | 115 | .86 | 228 | 228 | |
| -16 | 1.80 | 43 | 7.9 | 105 | .95 | 217 | 217 | |
| -17 | 2.20 | 45 | 7.9 | 95 | 1.10 | 202 | 202 | |
| -18 | 2.70 | 48 | 7.9 | 90 | 1.20 | 193 | 193 | |
| -19 | 3.30 | 49 | 7.9 | 80 | 1.30 | 185 | 185 | |
| -20 | 3.90 | 50 | 7.9 | 75 | 1.50 | 173 | 173 | |
| -21 | 4.70 | 50 | 7.9 | 70 | 2.40 | 136 | 136 | |
| -22 | 5.60 | 50 | 7.9 | 60 | 2.90 | 124 | 124 | |
| -23 | 6.80 | 50 | 7.9 | 55 | 3.20 | 118 | 118 | |
| -24 | 8.20 | 50 | 7.9 | 53 | 3.60 | 111 | 111 | |
| -25 | 10.0 | 50 | 7.9 | 50 | 4.00 | 106 | 106 | |
| -26 | 12.0 | 36 | 2.5 | 35 | 3.00 | 122 | 122 | |
| -27 | 15.0 | 38 | 2.5 | 30 | 3.40 | 115 | 115 | |
| -28 | 18.0 | 40 | 2.5 | 26 | 3.80 | 108 | 108 | |
| -29 | 22.0 | 40 | 2.5 | 24 | 4.90 | 96 | 96 | |
| -30 | 27.0 | 40 | 2.5 | 21 | 5.80 | 88 | 88 | |
| -31 | 33.0 | 41 | 2.5 | 20 | 6.50 | 83 | 83 | |
| -32 | 39.0 | 42 | 2.5 | 19 | 7.90 | 75 | 75 | 3/ |
| -33 | 47.0 | 44 | 2.5 | 16 | 9.30 | 69 | 69 | |
| -34 | 56.0 | 44 | 2.5 | 15 | 11.0 | 64 | 64 | |
| -35 | 68.0 | 45 | 2.5 | 13 | 12.0 | 61 | 61 | |
| -36 | 82.0 | 45 | 2.5 | 11 | 13.0 | 59 | 59 | |
| -37 | 100.0 | 40 | 2.5 | 10.5 | 16.8 | 51 | 51 | |

1/ Estimate only.
2/ Material for PIN's MIL-PRF-83446/34-01 through MIL-PRF-83446/34-25 is iron core-iron sleeve.
3/ Material for PIN's MIL-PRF-83446/34-26 through MIL-PRF-83446/34-37 is iron core-ferrite sleeve.



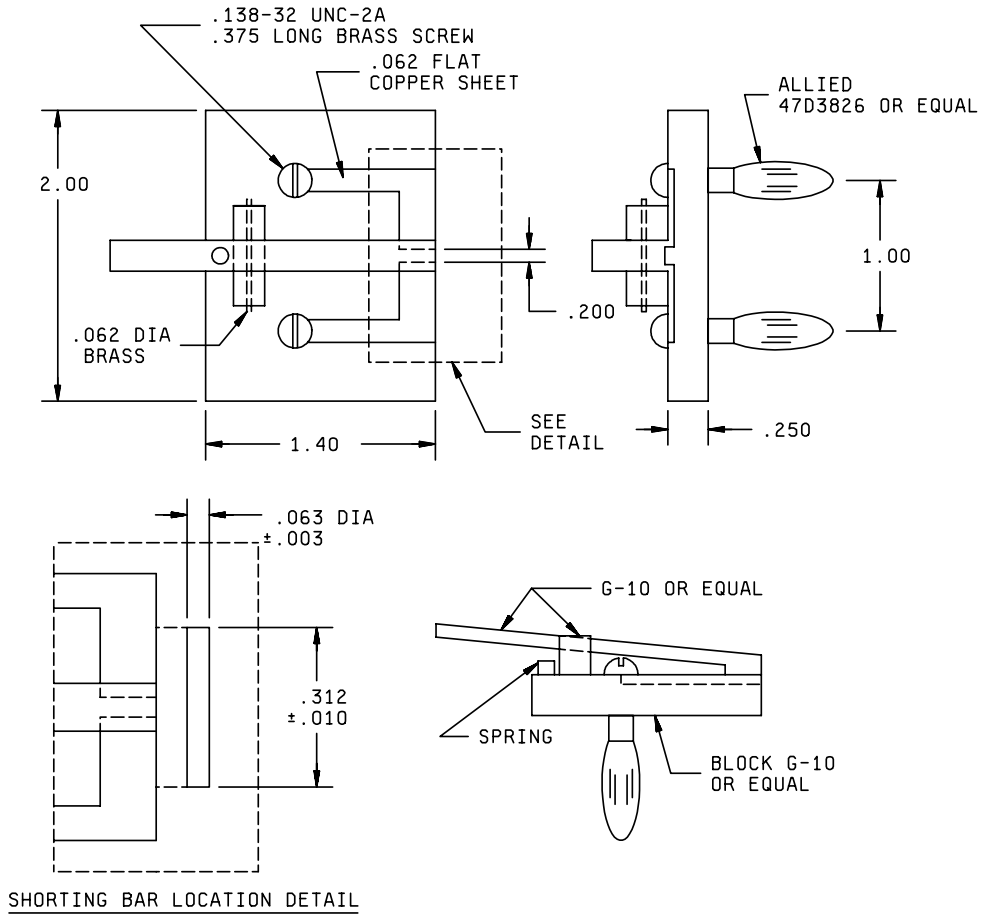
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TABLE II. Electrical characteristics (final).

| Inspection group | Allowable percent variation from the initial measurement | | | |
|---|---|---|---|---|
| Qualification Inspection: | Inductance | Q | Self-resonant Frequency | DC resistance |
| Group II | ±10 | -20 | -15 | ±3 (+.001 ohm) |
| Group IV | ±10 | -20 | -15 | ±3 (+.001 ohm) |
| Group V | ±10 | -20 | ----- | ----- |
| Quality Conformance Inspection: | | | | |
| Group C | | | | |
| Subgroup II | ±10 | -20 | -15 | ±3 (+.001 ohm) |
| Subgroup IV | ±10 | -20 | -15 | ±3 (+.001 ohm) |



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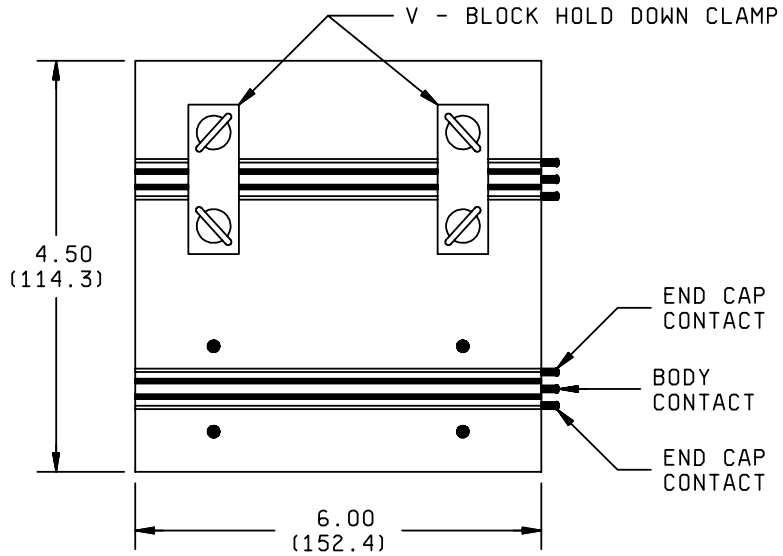


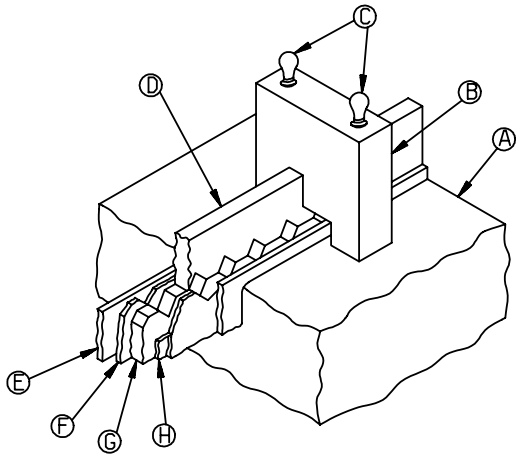

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Shorting bar material is brass.
4. The total fixture and residual inductance is approximately .049μH (for references only).

FIGURE 2. General test fixture, (TF-388)



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ENLARGED CUT-AWAY OF CLAMP

| Legend | |
|---|---|
| LTR | Description |
| A | G-10 base |
| B | Upper V-block hold-down clamp |
| C | Clamp screws |
| D | Upper V-block (body contact) |
| E | End cap contact retaining wall |
| F | End cap contact V-block (in contact with E) |
| G | Lower V-block (body contact) |
| H | G-10 insulators (separating G from F/E) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 3. Test fixture for insulation resistance, dielectric withstanding voltage and barometric pressure.



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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Bond strength: Force 2 pounds.

Weight: 0.30 gram maximum.

Operating temperature range: -55°C to +105°C.

Temperature rise (at +90°C): +15°C.

Maximum operating temperature: +105°C.

Altitude: 70,000 feet.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 200 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202, (70,000 feet), test voltage 80 volts rms.

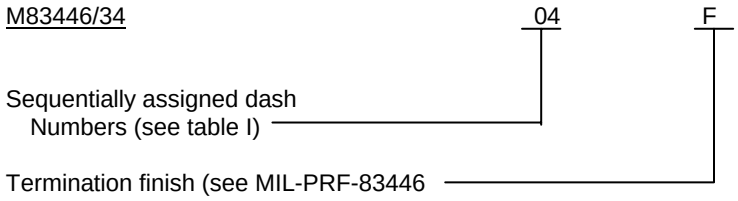
Electrical characteristics (initial): See table I.

Percent coupling: 3 percent maximum.

Electrical characteristics (final): See table II.

Test fixture: Tests shall be performed using test fixtures shown on figure 2 and 3, or equivalent.

Part or Identifying Number (PIN): The PIN shall be in the following form:




Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents.

MIL-PRF-83446
MIL-STD-202



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Army - CR
Navy - EC
Air Force - 11

Review activities:
Army - MI
Navy - AS, OS
Air Force – 19

Preparing activity:
DLA - CC

(Project 5950-1213)

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